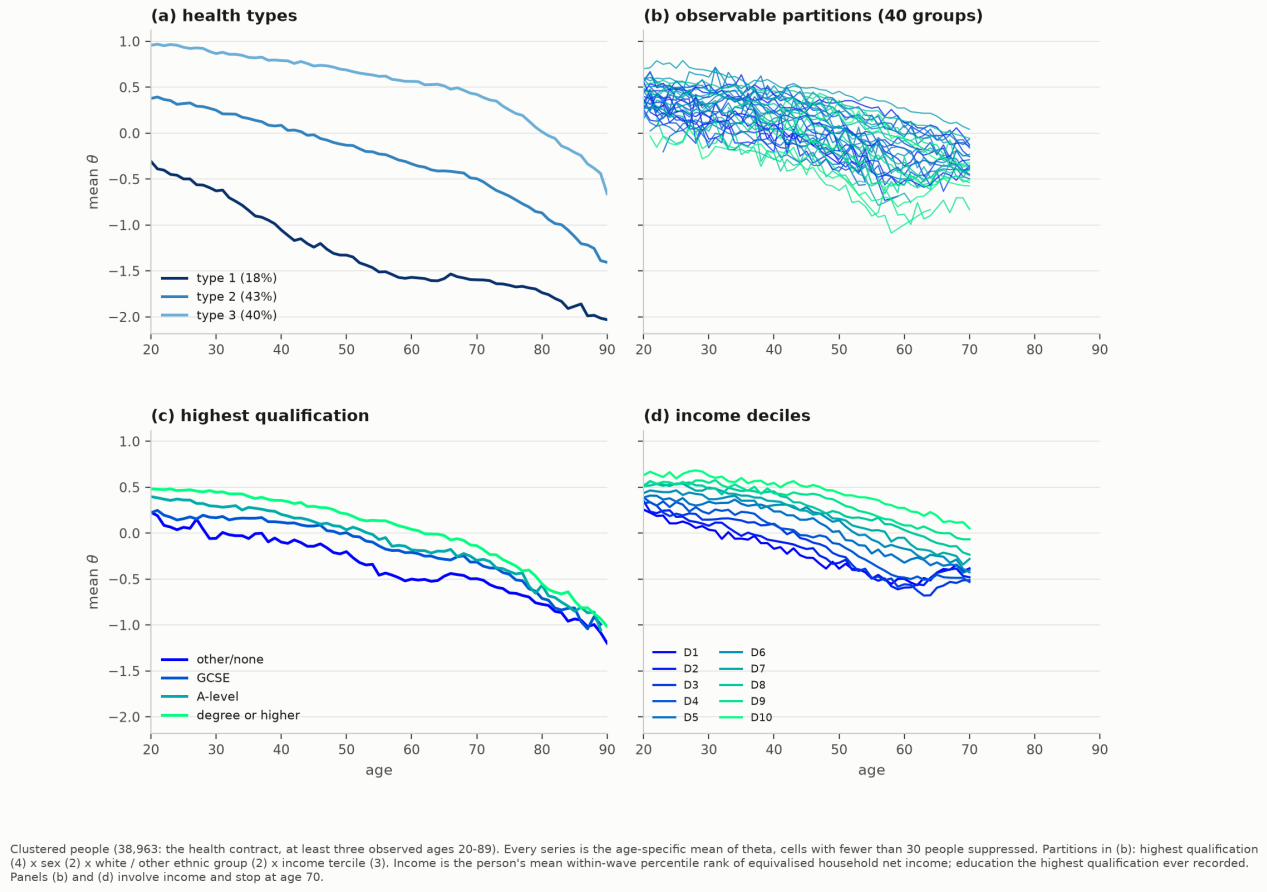


Figure 12: Lifecycle trajectories of θ by (a) health type, (b) 40 observable partitions, (c) education and (d) income decile, on the clustered people; cells with fewer than 30 people suppressed. The same figure on h is in Appendix C.

Predictive value.

- Rebuilt on the held-out fits (`descriptives/33_paper_prediction.py`): every Bayesian fit was refitted with each person's last two observed rows held out (the 29,901 people with at least five rows on the health contract; 26,664 people are in both the single-channel and the multidimensional held-out samples and the exercise runs on them). One prediction per person and horizon from the last fitted row t, with outcomes at the two held rows $t+1$ and $t+2$: the measure, the cost proxy in pounds (the flat-cost index capped at its 99th percentile, waves 7–15), and employment (employed or self-employed, people aged 20–64 at t). Class posteriors and the models' own forecasts are computed from the fitted window alone and carried to the outcome's age, so nothing after t informs them. People are split 70/30, fitted on the 70 and scored on the 30 (Figure 13, Table 3).
- **Observables predict almost nothing about health or cost, but more than health about employment.** Education, income and sex add 0.06 of R^2 for θ_{t+1} and 0.01 for cost over age, against 0.48–0.54 and 0.08–0.09 for the held-out classes. For employment they lift the AUC from 0.64 to 0.78, against 0.74–0.76 for any health summary, and the two together reach 0.82. Employment at t alone reaches 0.90: once current status is known, health adds little to the labour-market outcome.
- **The current score and the classes carry about the same information, and the model's**

forecast beats both. θ_t alone reaches $R^2 = 0.69$ for θ_{t+1} and 0.14 for cost, against 0.68 and 0.12 for the AR(1) plus "spike" classes, and adding the classes to the score gains 0.03 for health and nothing for cost. The model's own forecast of θ_{t+1}, the class-weighted path plus the filtered state carried forward, one number per person, does better than the score plus its lag ($R^2 = 0.74$ against 0.73): the classes summarise the history, and the history is what matters.

- **Two waves ahead the ranking is unchanged.** R^2 for θ_{t+2} is 0.61 for the iid classes, 0.66 for the AR(1) plus "spike" classes, 0.66 for θ_t and 0.71 for the model forecast, against 0.62, 0.68, 0.69 and 0.74 one wave ahead; for cost the score falls from 0.14 to 0.11 while the classes hold at 0.12. The classes lose less over the extra wave than the current score does and draw level with it at $t + 2$, but only the forecast, which carries the filtered state forward, stays clear.
- The multidimensional classes and forecast match the single-channel ones (R^2 0.61 and 0.74 for θ_{t+1} against 0.68 and 0.74; 0.11 for cost against 0.12): the mental channel and the mortality hazard neither help nor hurt the physical forecast. The same holds on h (0.63 and 0.77 against 0.68 and 0.78; Figure C.18).

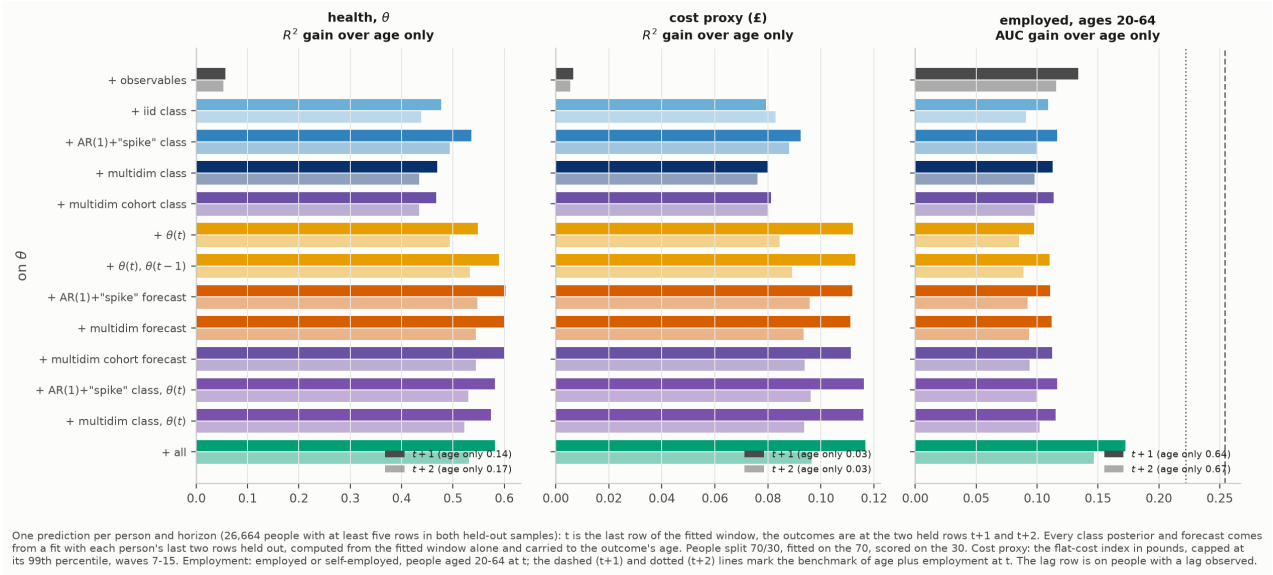


Figure 13: Out-of-sample gain over an age-only quadratic in predicting the measure, the cost proxy in pounds and employment (ages 20–64) one and two waves ahead, from each set of predictors known at t, on θ, with every class posterior and forecast from a fit that held out the two target rows; the cohort rows use the multidimensional fit with birth-decade shifts. In the employment panel the dashed $(t+1)$ and dotted $(t+2)$ lines mark age plus employment at t. The same figure on h is in Appendix C.

predictors at t	health, measure (R^2)		cost proxy (£) (R^2)		employed, ages 20-64 (AUC)	
	$t+1$	$t+2$	$t+1$	$t+2$	$t+1$	$t+2$
on θ						
age	0.140	0.167	0.032	0.027	0.643	0.666
+ observables	0.197	0.219	0.039	0.033	0.777	0.782
+ iid class	0.618	0.605	0.111	0.110	0.753	0.758
+ AR(1)+"spike" class	0.677	0.661	0.124	0.115	0.760	0.766
+ multidim class	0.609	0.601	0.112	0.103	0.756	0.764
+ multidim cohort class	0.608	0.601	0.113	0.107	0.757	0.764
+ theta(t)	0.689	0.661	0.144	0.112	0.741	0.752
+ theta(t), theta(t-1)	0.730	0.699	0.145	0.116	0.754	0.755
+ AR(1)+"spike" forecast	0.743	0.714	0.144	0.123	0.754	0.759
+ multidim forecast	0.740	0.711	0.143	0.121	0.755	0.760
+ multidim cohort forecast	0.741	0.711	0.143	0.121	0.756	0.760
+ AR(1)+"spike" class, theta(t)	0.722	0.697	0.148	0.123	0.760	0.767
+ multidim class, theta(t)	0.715	0.689	0.148	0.121	0.759	0.768
+ all	0.723	0.697	0.148	0.124	0.816	0.813
benchmark: age, employed(t)					0.897	0.889

Table 3: Out-of-sample scores by predictor set, outcome and horizon, on the held-out fits on θ; R^2 for the continuous outcomes, AUC for employment. The same table on h is in Appendix C.

Predicting death.

- The held-out design cannot predict death at a fixed horizon: everyone at t has two more rows, and $t+2$ is by construction everyone's last interview. So mortality is shown two ways (Figure 14, `descriptives/41_paper_mortality_prediction.py`). On the full-sample fits, death before the next wave at every person-wave at risk (308,949 person-waves, 1,267 deaths): the single-channel classes never saw the death record, and the multidimensional class is taken from the physical and mental channels only, with the full-likelihood version, which has seen the decedent's event, shown as an upper bound. On the held-out fits, death after the last interview against leaving the panel alive, predicted from t (13,516 people, 7% deaths), with the multidimensional posterior from survival through the fitted window only.
- **Death is the outcome health predicts least beyond age.** On θ, at every person-wave, age alone gives AUC 0.85; any class adds about 0.03 (0.88), the current score 0.05 (0.90), and the classes add nothing on top of the score. The multidimensional class from the health channels alone does as well as the single-channel classes, and so does its own class-weighted Gompertz–Makeham hazard, one number per person; letting the class see the death record buys about 0.01. On the held-out comparison the ordering is the same, from 0.80 for age to 0.83 for the classes or the score. Utilisation is the opposite case: health does nearly all the work there, so prevention targeted on these measures should expect to move costs far more readily than mortality.

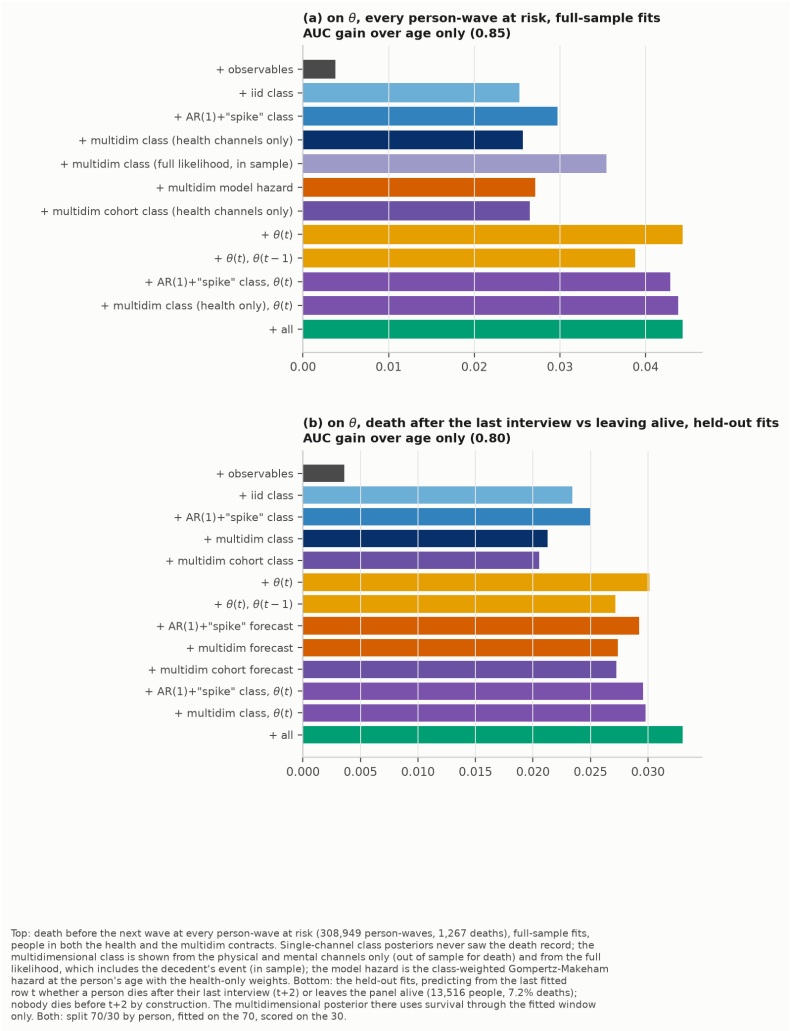


Figure 14: Predicting death, AUC gain over an age-only quadratic. (a, c) Every person-wave at risk on the full-sample fits: death before the next wave. (b, d) The held-out fits: death after the last interview against leaving the panel alive, predicted from the last fitted row. The cohort rows use the multidimensional fit with birth-decade shifts. On θ; the same figure on h is in Appendix C.



Figure 15: What identifies the type: the observables or a health reading? On θ; every typology is fitted without each person's first two rows (people with at least five rows, 29,900): partial K-means on the remaining rows, and the independent-residuals and AR(1) plus "spike" mixtures, without and with birth-decade shifts, with the first two rows held out, the type being the modal class of the posterior from the fitted window. The multidimensional AR(1) plus "spike" fit with mortality is likewise refitted with the first two rows held out (on the multidimensional contract's people with at least five rows, 29,278). Each predictor set is known at the first row and sits on a quadratic in age: education, income rank, the three observables together, one reading of θ, the two held readings, and the first reading plus the observables; multinomial logit, people split 70/30, scored on the 30. (a) Accuracy, dashed the majority-type rate; (b) macro one-vs-rest AUC. "Model posterior": the class posterior the fitted mixture itself implies from the first observation, or from both held rows, no regression.

predictors at the first rows	K-means		indep. residuals		AR(1) + "spike"		+ cohort shifts		multidim	
	accuracy	AUC	accuracy	AUC	accuracy	AUC	accuracy	AUC	accuracy	AUC
majority type	0.424		0.436		0.587		0.561		0.525	
age(0)	0.429	0.535	0.443	0.530	0.587	0.589	0.561	0.549	0.525	0.533
+ education	0.457	0.585	0.469	0.584	0.587	0.627	0.561	0.596	0.525	0.577
+ income rank	0.468	0.615	0.490	0.616	0.593	0.646	0.563	0.621	0.529	0.606
+ education, income, sex	0.469	0.625	0.489	0.626	0.593	0.655	0.563	0.631	0.532	0.619
+ $\theta(0)$	0.640	0.798	0.651	0.802	0.692	0.805	0.670	0.800	0.630	0.772
+ $\theta(0)$, $\theta(1)$	0.693	0.847	0.698	0.853	0.731	0.849	0.716	0.847	0.670	0.813
+ $\theta(0)$, education, income, sex	0.645	0.805	0.656	0.809	0.694	0.811	0.680	0.806	0.633	0.780
+ $\theta(0)$, mental $\theta(0)$									0.669	0.810
model posterior from the first row			0.637	0.800	0.687	0.787	0.669	0.793	0.650	0.807
model posterior from $\theta(0)$, $\theta(1)$			0.696	0.849	0.730	0.833	0.716	0.840		

Table 4: Accuracy and macro one-vs-rest AUC on the test 30% for each predictor set and typology, Figure 15.

- **What identifies the type: a health reading, or education and income?** (Figure 15, Table 4; `descriptives/43_paper_type_from_initial.py`). The types are refitted without each person's first two rows, so that the first two observations can be used to predict them without having shaped

them (the class being identified from the fitted window, as in Figure 10), and the observables are set against one reading of θ at the first row and against the two readings together.

- **The observables say little.** On the K-means types, age alone matches the majority rate (0.43 against 0.42, AUC 0.54); education takes the AUC to 0.59, income rank to 0.62, and all three observables together to 0.63 with accuracy 0.47. The same holds on every typology: the observables add 0.08–0.10 of AUC over age and 0–5 points of accuracy.
- **One health reading says a lot.** θ at the first row takes the AUC to 0.80–0.81 on every single-channel typology and the accuracy to 0.64–0.69, two people in three. On top of the reading the observables add one point of accuracy and 0.01 of AUC: nearly everything education and income know about the type, the health reading already knows.
- **A second reading, a year or so later, is worth more than all the observables.** The two held readings together take the AUC to 0.85 on every typology and add four to five points of accuracy (0.69–0.73), the gain from averaging out one "spike". So the ranking is observables, then one reading, then two readings, with the first step the smallest.
- **The fitted model is the right classifier.** The mixture's own posterior from the first observation, which discounts it for the "spike", does as well as the regression on one reading (0.69 against 0.69 on the AR(1) plus "spike" classes; 0.64 against 0.65 on the independent-residuals classes), and its posterior from both held rows does as well as the regression on two (0.73 against 0.73; 0.70 against 0.70). On the AR(1) plus "spike" classes the majority rate is 0.59 because the posterior concentrates its modal assignments in the middle class (59% modal against a 48% share), so the accuracy gain is smaller there while the AUC is the same. The typology with birth-decade shifts behaves like the one without.
- On the multidimensional classes, θ at the first row gives AUC 0.77; a second reading of θ takes it to 0.81, and so does the mental score at the first row (0.81), as does the model's own posterior from the two first-row scores. A second reading of the physical channel and one reading of the other channel are worth the same to the joint type, and the observables again add 0.01. *[The two-row model posterior is implemented for the single-channel mixtures only; the multidimensional row's model posterior uses the first row.]*
- A reading at 20–34 classifies the K-means and independent-residuals types worse (0.60–0.61) than one at 35 or later (0.65–0.68). On h (Figure C.21) the picture is the same, with two differences: age alone does more (0.58–0.61 of AUC), because on the bounded scale the young sit at the ceiling, and the AR(1) plus "spike" model's own posterior falls short of the regression by 0.04–0.05 of AUC, the Gaussian observation model misfitting a bounded, skewed scale.

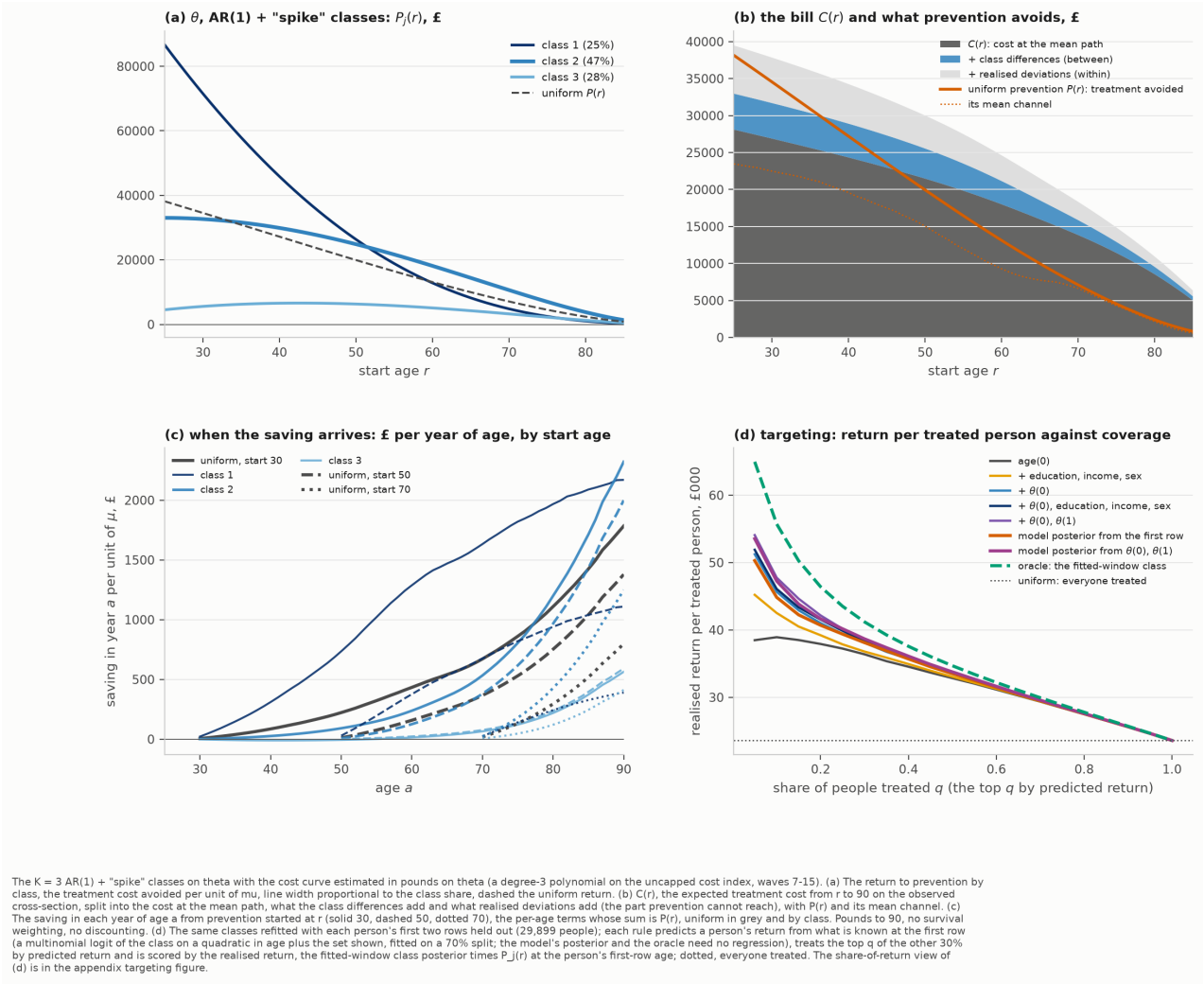
5 Prevention exercises

We price the returns of Section 2 on the $K = 3$ AR(1) plus "spike" classes on θ (Figure 6(b)), with the independent-residuals classes alongside for comparison. The cost function is the curve estimated in pounds on θ (Figure 1), a cubic fitted to the uncapped cost index over waves 7–15, and the returns take the general-cost form of Section 2.2. Class shares and paths are the fitted mixture at its posterior mean; $\mathbb{E}[-c'(\theta) \mid j]$ at each age is the posterior-weighted mean over the person-waves observed there, and the bill $C(r)$ and its three parts are the observed cross-section priced by the same curve. Returns are pounds of treatment cost to 90 avoided per unit of μ, with no survival weighting and no discounting. The K-means version and the capped cost curve are in Appendix G; the same exercise on h is in Appendix C.

- **The bill and what prevention avoids** (Figure 16(a), (b); Table 5). The treatment bill from 30

is £38,000 per person to 90, £30,000 from 50 and £18,000 from 70. On the AR(1) plus "spike" classes about 71% of it is the cost along the mean path, 13% is what the type differences add and 16% is realised deviations, the part no prevention reaches (15% from 50, 13% from 70). Uniform prevention from 30 avoids £35,000 of it per unit of μ, so $P(r)/C(r)$ is 0.91 at 30, falling to 0.67 at 50 and 0.39 at 70: the later the start, the larger the share of the remaining bill that is already determined. On the independent-residuals classes the shares are 0.68, 0.47 and 0.26, since more of the decline is read as noise.

- **Gap times preventable decline** (Result 2). Panels (a) and (c) of Figure 16 are the two factors of (5) multiplied age by age: (c) is $s_j(a \mid r)$, the gap at a times the decline opened by a, and (a) is its sum over ages. From 30 the worst class returns twice the uniform return (£71,500 against £34,500) because its gap and its decline coincide, already below the others at 30 and falling fastest; the middle class returns about the uniform amount; the best class, healthy and nearly flat, a sixth of it (£5,600). The mean channel of Corollary 1 is 65% of the uniform return from 30, 75% from 50 and 93% from 70, the between-class covariance the rest.
- **Timing.** On the AR(1) plus "spike" classes the between-class channel of $P(r)$ is positive at every start age: +35% of $P(30)$, +25% of $P(50)$ and +7% of $P(70)$. On the independent-residuals classes it is +13% at 30 and turns negative by 50 (−8%), reaching −42% at 70: the sign change of Figure 2 priced. Once the one-period noise is absorbed, the classes fan out for longer, and divergence keeps working for uniform prevention into old age.
- **Targeting on the class.** From 30 the worst class's return is £72,000, against £35,000 uniform and £5,600 for the best class, so class targeting collects about twice the uniform return (a gain of 107%). From 50 the worst class still has the highest return and the gain is 32%; from 70 the middle class does, and the gain is 50%. On the independent-residuals classes the gains are 151%, 49% and 50%.
- Skeleton for the full version:
  - posterior uncertainty, from the draws;
  - survival-weighted sums $\sum_{a \geq r} S(a \mid r)(\cdot)$ using the Gompertz–Makeham class hazards of the multidimensional model, both ways (the sickest die, prevention raises survival);
  - targeting on observables versus on early health history versus on the class, priced by the targeting identity of Section 2, using the Section 4.4 partitions;
  - the additive-prevention robustness check (the multiplicative $(1 - \mu)d$ favours decliners by construction), and prevention as moving a person to a better class;
  - cost sides: the returns are pounds of treatment avoided per unit of μ, so a policy's value still needs the cost of delivering a unit of μ; and when the savings come, since prevention is paid at r and the avoided treatment falls decades later.



The K = 3 AR(1) + "spike" classes on theta with the cost curve estimated in pounds on theta (a degree-3 polynomial on the uncapped cost index, waves 7-15). (a) The return to prevention by class, the treatment cost avoided per unit of mu, line width proportional to the class share, dashed the uniform return. (b) C(r), the expected treatment cost from r to 90 on the observed cross-section, split into the cost at the mean path, what the class differences add and what realised deviations add (the part prevention cannot reach), with P(r) and its mean channel. (c) The saving in each year of age a from prevention started at r (solid 30, dashed 50, dotted 70), the per-age terms whose sum is P(r), uniform in grey and by class. Pounds to 90, no survival weighting, no discounting. (d) The same classes refitted with each person's first two rows held out (29,899 people); each rule predicts a person's return from what is known at the first row (a multinomial logit of the class on a quadratic in age plus the set shown, fitted on a 70% split; the model's posterior and the oracle need no regression), treats the top q of the other 30% by predicted return and is scored by the realised return, the fitted-window class posterior times P_j(r) at the person's first-row age; dotted, everyone treated. The share-of-return view of (d) is in the appendix targeting figure.

Figure 16: Prevention on the $K = 3$ AR(1) plus "spike" classes on θ, in pounds under the estimated cost curve. (a) The return to prevention by class, $P_j(r)$, the treatment cost avoided per unit of μ by start age r, with the uniform return dashed. (b) The treatment bill still to come $C(r)$, stacked into the cost at the mean path, what the class differences add and what realised deviations add, with $P(r)$ and its mean channel. (c) When the saving arrives: the saving in each year of age a from prevention started at 30 (solid), 50 (dashed) and 70 (dotted), the per-age terms whose sum is $P(r)$, uniform in grey and by class. (d) Targeting, on the same classes refitted with each person's first two rows held out: the realised return per treated person when the top share q by predicted return is treated, by what the prediction uses (age; education, income and sex; one reading of θ; the reading and the observables; two readings; the model's own posterior from one row or from two; the oracle that knows the class), dotted everyone treated. The cumulative view of (c) with the present values, and the share-of-return view of (d), are in Appendix G.2.

classes	r	$C(r)$	within	$P(r)$	P/C	between	P_1	P_2	P_3	gain
AR(1) + "spike" classes	30	37,834	16%	34,542	0.91	+35%	71,536	32,653	5,580	1.07
AR(1) + "spike" classes	50	30,005	15%	19,953	0.66	+25%	26,356	24,866	6,314	0.32
AR(1) + "spike" classes	70	18,339	13%	7,111	0.39	+7%	4,828	10,650	3,265	0.50
independent-residuals classes	30	37,834	11%	25,840	0.68	+13%	64,804	28,832	8,076	1.51
independent-residuals classes	50	30,005	10%	13,972	0.47	-8%	20,869	18,613	6,443	0.49
independent-residuals classes	70	18,339	9%	4,679	0.26	-42%	2,872	7,001	2,879	0.50

Table 5: The bill and the returns at three start ages on θ, in pounds under the estimated cost curve. $C(r)$ is the treatment bill still to come and "within" the share of it that realised deviations add, the part prevention cannot reach; $P(r)$ is the treatment cost uniform prevention avoids per unit of μ and P/C its share of the bill; "between" is the between-class channel of $P(r)$ as a share of $P(r)$; P_j the return by class; the gain is $(\max_j P_j - P)/P$.

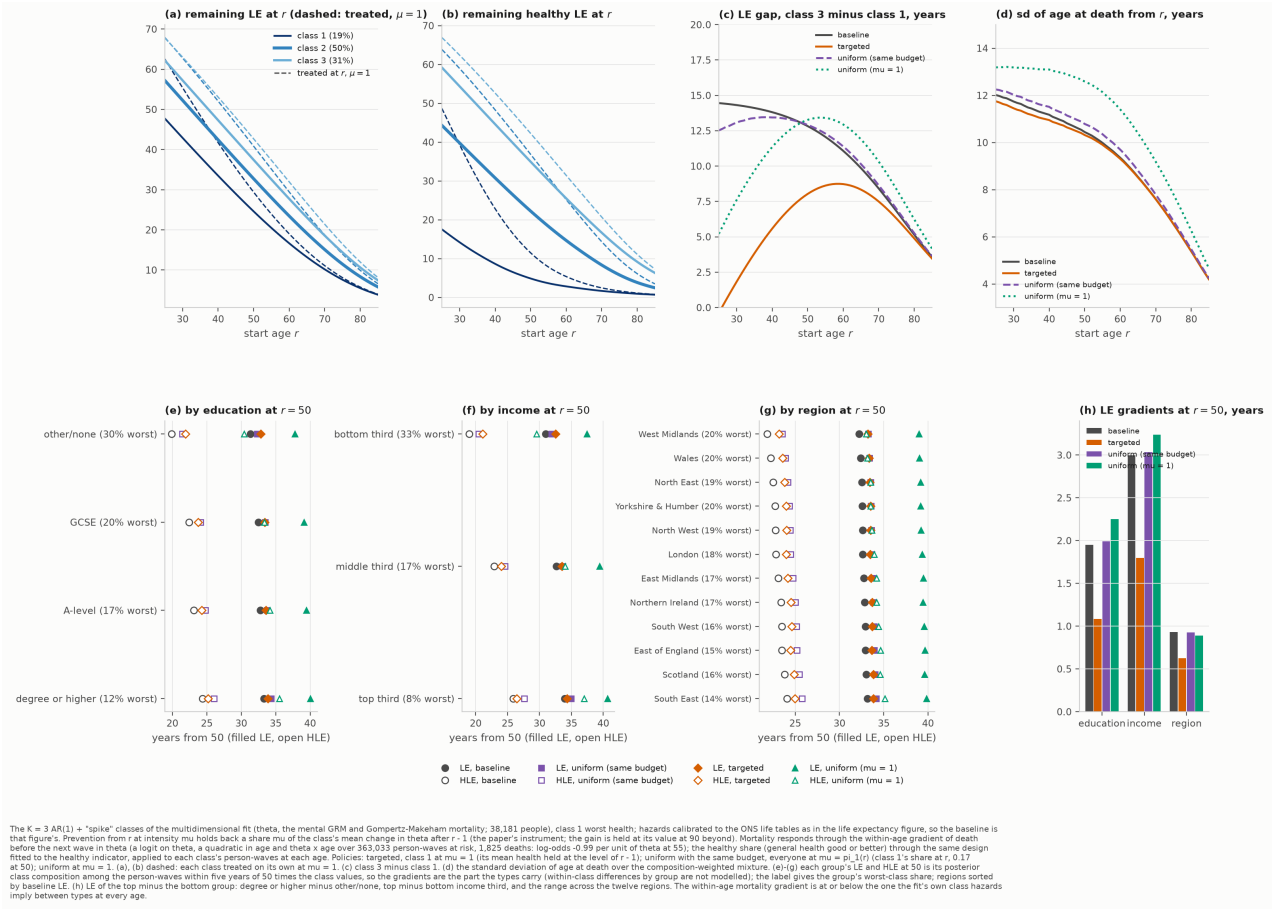
targeting on	top 10% treated			top 25% treated			top 50% treated		
	£	/uniform	/oracle	£	/uniform	/oracle	£	/uniform	/oracle
age(0)	38,923	1.65	0.70	37,236	1.58	0.86	32,847	1.39	0.95
+ education, income, sex	42,515	1.80	0.76	37,860	1.60	0.87	33,053	1.40	0.95
+ $\theta(0)$	45,654	1.94	0.82	39,427	1.67	0.91	33,522	1.42	0.97
+ $\theta(0)$, education, income, sex	46,093	1.95	0.83	39,876	1.69	0.92	33,626	1.43	0.97
+ $\theta(0)$, $\theta(1)$	47,763	2.02	0.86	40,167	1.70	0.92	33,758	1.43	0.97
model posterior from the first row	44,842	1.90	0.80	39,386	1.67	0.91	33,520	1.42	0.97
model posterior from $\theta(0)$, $\theta(1)$	47,253	2.00	0.85	40,169	1.70	0.92	33,773	1.43	0.97
oracle: the fitted-window class	55,731	2.36	1.00	43,517	1.84	1.00	34,687	1.47	1.00

Table 6: Realised return per treated person, in pounds of treatment avoided per unit of μ, when the top 10%, 25% or 50% by predicted return are treated, by targeting rule, on the classes refitted with the first two rows held out (29,900 people); as a multiple of the uniform return (everyone treated, £23,600) and as a share of the oracle's.

- **When the saving arrives** (Figure 16(c); Figure G.37 in Appendix G.2 for the cumulative view). $P_j(r)$ is a sum over ages of the saving in each year, $E[-c'(h_a) \mid j](\bar{h}_{j,r-1} - \bar{h}_{j,a})$: the gap the policy has opened by age a times the marginal cost of a gap at that age, both of which grow with age. So prevention is paid now and almost nothing comes back for decades. Started at 30, 1% of the uniform return has arrived by 40 and half only by 78; started at 50, half by 82; started at 70, half by 85. The worst class pays back soonest (half by 73 from a start at 30), the middle class latest (82), because its decline comes late. The saving per year is still rising at 90, so the horizon matters too.
- **Discounting therefore bites hard, but does not reverse the ranking of start ages.** At the Green Book 3.5%, the present value of the uniform return is 24% of its undiscounted value from 30, 38% from 50 and 62% from 70: about £8,300, £7,600 and £4,400 per unit of μ against £35,000, £20,000 and £7,100 undiscounted. Early prevention still returns the most in present value, but by a fifth rather than a factor of five, and the comparison with a prevention cost paid at r is the economic question this sets up. [**TODO:** Add the cost of a unit of μ and the survival weighting; both change the timing arithmetic.]
- **What to target on** (Figure 16(d); Table 6; `descriptives/46_paper_targeting.py`). The class returns of Table 5 assume the class is known. It is not: a policy can target only on what it sees. Pricing rules on what is known at a person's first observations, treating the top tenth by predicted return collects £38,900 per treated person on age alone (returns fall with age, so the young come

first), £42,500 adding education, income and sex, £45,700 adding one reading of θ, £46,100 with the reading and the observables, and £47,800 with two readings, against £55,700 for the oracle that knows the fitted class and £23,600 for treating everyone.

- **The health reading is the information that matters, and it is already most of what there is.** Of the gap between the age-only rule and the oracle at 10% coverage, the observables close a fifth, one reading of θ closes 40%, the observables add two points on top of the reading, and a second reading takes it past half (53%). The mixture's own posterior does as well as the regression on the same rows (£44,800 from one, £47,300 from two). At 25% coverage the ordering is the same and the gaps halve (£37,200, £37,900, £39,400, £40,200 against £43,500); by half coverage every rule is within 5% of the oracle, because the oracle's advantage is in finding the small worst class.
- On h (Figure C.22) the age-only rule is far weaker (£16,700 at 10% coverage against £31,900 with one reading, £34,300 with two and £43,600 for the oracle), because on the bounded scale the young and healthy are indistinguishable and the worst class is smaller; the observables matter more there (£24,800), but still less than one reading.
- The remaining gap to the oracle is what a longer health history buys: the class is identified from a person's whole fitted window, and two readings recover about half of the return it would collect. [**TODO:** Price the rules net of a cost of screening, and add targeting on the predicted class rather than on the predicted return (which mixes age and type).]



The K = 3 AR(1) + "spike" classes of the multidimensional fit (theta, the mental GRM and Gompertz-Makeham mortality; 38,181 people), class 1 worst health; hazards calibrated to the ONS life tables as in the life expectancy figure, so the baseline is that figure's. Prevention from r at intensity mu holds back a share mu of the class's mean change in theta after r - 1 (the paper's instrument; the gain is held at its value at 90 beyond). Mortality responds through the within-age gradient of death before the next wave in theta (a logit on theta, a quadratic in age and theta x age over 363,033 person-waves at risk, 1,825 deaths: log-odds -0.99 per unit of theta at 55); the healthy share (general health good or better) through the same design fitted to the healthy indicator, applied to each class's person-waves at each age. Policies: targeted, class 1 at mu = 1 (its mean health held at the level of r - 1); uniform with the same budget, everyone at mu = pi_1(r) (class 1's share at r, 0.17 at 50); uniform at mu = 1. (a), (b) dashed: each class treated on its own at mu = 1. (c) class 3 minus class 1. (d) the standard deviation of age at death over the composition-weighted mixture. (e)-(g) each group's LE and HLE at 50 is its posterior class composition among the person-waves within five years of 50 times the class values, so the gradients are the part the types carry (within-class differences by group are not modelled); the label gives the group's worst-class share; regions sorted by baseline LE. (h) LE of the top minus the bottom group: degree or higher minus other/none, top minus bottom income third, and the range across the twelve regions. The within-age mortality gradient is at or below the one the fit's own class hazards imply between types at every age.

Figure 17: What targeted prevention does to life expectancy and to inequality. The $K = 3$ AR(1) plus "spike" classes of the multidimensional fit with their calibrated hazards (Figure 11); prevention from r at intensity μ holds back a share μ of the class's mean change in θ after $r - 1$, mortality responds through the within-age gradient of death in θ and the healthy share through the same gradient in the healthy indicator. Policies: targeted, the worst class at $\mu = 1$; uniform with the same budget, everyone at $\mu = \pi_1(r)$; uniform at $\mu = 1$. (a), (b) Remaining LE and HLE by class, and each class if treated on its own at $\mu = 1$ (dashed). (c) The LE gap between the best and the worst class. (d) The standard deviation of age at death over the population mixture. (e)–(g) LE (filled) and HLE (open) from 50 by education, income third and region, each group's class composition among the person-waves within five years of 50 times the class values, with the group's worst-class share in the label. (h) The LE gradients: degree or higher minus other/none, top minus bottom income third, the range across regions.

		population		inequality			gradient in LE, years		
from	policy	LE	HLE	sd(death age)	LE gap	HLE gap	education	income	region
30	none	51.7	38.6	11.7	14.3	40.2	2.2	2.6	1.2
	worst class, $\mu = 1$	54.3	43.7	11.5	1.8	15.1	-0.0	0.2	0.4
	everyone, $\mu = 0.20$	53.5	41.7	12.0	13.1	38.5	2.1	2.4	1.1
	everyone, $\mu = 1$	61.3	55.9	13.2	7.6	23.1	1.4	1.4	0.5
50	none	32.8	23.2	10.4	12.8	30.1	2.0	3.0	0.9
	worst class, $\mu = 1$	33.6	24.3	10.3	8.0	23.5	1.1	1.8	0.6
	everyone, $\mu = 0.17$	33.8	24.8	10.8	12.9	30.6	2.0	3.0	0.9
	everyone, $\mu = 1$	39.5	34.2	12.6	13.3	30.8	2.3	3.2	0.9
70	none	15.9	10.8	7.6	8.4	15.0	1.0	1.1	0.8
	worst class, $\mu = 1$	16.0	10.9	7.6	7.5	14.2	0.9	1.0	0.7
	everyone, $\mu = 0.13$	16.2	11.3	7.8	8.6	15.4	1.0	1.1	0.8
	everyone, $\mu = 1$	18.8	15.3	9.2	10.3	18.3	1.1	1.2	1.1

Table 7: Population LE and HLE from r, the standard deviation of age at death, the class-3-minus-class-1 gaps in LE and HLE, and the LE gradients (education: degree or higher minus other/none; income: top minus bottom third; region: range), in years, under no prevention and the three policies of Figure 17; the uniform "same budget" intensity is the worst class's share at r.

- **Life expectancy, and who gains** (Figure 17(a), (b); Table 7). The returns above are pounds; this prices the same instrument in years, on the multidimensional fit's classes and calibrated hazards of Figure 11, so the baseline is Table 2 (`descriptives/48_paper_prevention_le.py`); the same exercise on the univariate AR(1) plus "spike" classes of Figure 16, with class hazards estimated for them, is Figure G.40 in Appendix G.3. The link from health to death is the within-age gradient of mortality in θ (-1.0 log-odds per unit at 55, from the logit of Section 3), which is at or below the gradient the fit's own class hazards imply between types at every age, so the years are if anything conservative. Holding a class's mean health at its level at 49 ($\mu = 1$ from 50) adds 4.8 years of LE for the worst class, 8.1 for the middle and 5.3 for the best, and more HLE than LE in every class (6.6, 14.6 and 7.3 years), since held-back decline keeps people on the "good" side of the self-rated line for longer than it keeps them alive.
- **Targeting the worst type narrows the gap between types; the same budget spread evenly does not.** From 50 the LE gap between the best and the worst class is 12.8 years. Treating the worst class at $\mu = 1$ takes it to 8.0 (and the HLE gap from 30 to 24); the same budget spread evenly ($\mu = 0.17$ for everyone) leaves both gaps where they are, because every class gains in proportion and the worst class's gain is not larger than the middle class's. Population LE rises by 0.8 years under targeting and 1.0 under the even spread: the even spread buys slightly more years in total, the targeted policy buys them where they close a gap. From 30 the targeted policy is extreme, since it holds the worst class flat for sixty years: its LE rises by 12.5 years to above the middle class's, and the type gap closes to 1.8 years. From 70 neither policy does much (0.1–0.3 years), as the decline that prevention can reach is mostly behind.
- **Lifespan inequality is mostly within type, and prevention does not reduce it.** The standard deviation of age at death from 50 is 10.4 years at baseline, 10.3 under targeting and 10.8 under the even spread; from 30, 11.7, 11.5 and 12.0. The gap between types is a small part of the spread in ages at death, which is dominated by the Gompertz randomness within a type, and any policy that lengthens lives spreads them out. Inequality between types, not inequality in lifespans, is what targeted prevention moves.
- **The gradients the types carry are small, and targeting halves them.** Worst-class membership runs from 30% among people with no qualifications to 12% among graduates, from 33% in the bottom income third to 8% in the top, and between 14% and 20% across regions.

Through that composition the types carry an LE gradient from 50 of 2.0 years by education, 3.0 by income and 0.9 across regions (HLE: 4.4, 6.9 and 2.2); the published UK gradients are several times larger, so the types carry only part of them, and the rest is within type. Treating the worst class from 50 cuts the education gradient to 1.1 years, the income gradient to 1.8 and the regional range to 0.6, while the same budget spread evenly leaves all three unchanged; from 30 the targeted policy removes the education and income gradients the types carry altogether. [**TODO:** The gradients here are the composition channel only; add the within-class gradients (class-specific LE by education and income, or a Gompertz fit by group) to say what share of the published gradients runs through the types. Also the cost side: a unit of μ for the worst class may cost more than for the others.]

Appendices

A Proofs and additional material for conceptual framework

[Include proofs for any propositions and perhaps some additional illustrative figures here.]

B Health measure and cost proxy

B.1 Health variables

Six SF-12 items enter as four testlets: the two physical-function items sum into PF, the two role-physical items sum into RP, and general health and pain enter alone. *[add explanation of relationship to PCS here.]* Every item is coded so category 1 is worst health. In addition to the SF-12 items, UKHLS asks which of a list of areas a person's health limits them in. Eight of these are physical and used here, in three groups:

1. LIM_FL (0-4): mobility (`disdif1`), lifting (`disdif2`), dexterity (`disdif3`), co-ordination (`disdif10`
2. LIM_SELF (0-2): continence (`disdif4`), personal care (`disdif11`)
3. LIM_SENS (0-2): hearing (`disdif5`), sight (`disdif6`)

In the early waves the battery was asked only of people reporting a long-standing illness. Someone who reports none therefore has no answer, not a zero. We count them as having no limitation, in every wave, so that the variable means the same thing across the panel; someone who does report a long-standing illness but has no usable answer is missing. Two areas are deliberately left out. "Recognising when in physical danger" is included with the mental health items and "Other health problem or disability" is included as a robustness variant and changes nothing. COND counts the sixteen ever-diagnosed physical conditions in the UKHLS inventory, 0 to 5 and 6 or more, the top category being the highest count whose share of person-waves is at least 0.5% (6 or more holds 0.86%).

Table B.1 shows the unconditional distribution of health variables. In total, 7.1% of person-waves give the best answer to all eight items, while only 12 person-waves sit at the worst answer on everything.